WORKED SOLUTION

STEP III 2021, Question 3

Pure • Integral Inequalities

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Differentiate the thing being raised to the power n. For $u = \sec x + \tan x$ the derivative is $u \sec x$, which makes $u^{n-1}\frac{du}{dx}$ exactly integrable, so the job is to rewrite the integrand of $I_{n+1} + I_{n-1}$ in that form. The inequalities then come from comparing the integrand $u^n = u^{n-1} \cdot u$ with the exactly integrable $u^{n-1} \cdot u'$: if $u < u'$, the integral is less than $[u^n/n]$.

Solution

Part (i)

Let $u = \sec x + \tan x$. Then

$$\frac{du}{dx} = \sec x \tan x + \sec^2 x = \sec x\,(\sec x + \tan x) = u \sec x.$$

Also, using $\tan^2 x + 1 = \sec^2 x$,

$$u^2 + 1 = \sec^2 x + 2\sec x \tan x + \tan^2 x + 1 = 2\sec^2 x + 2\sec x \tan x = 2u \sec x = 2\frac{du}{dx}.$$

So for $n \geq 1$,

$$\begin{aligned} I_{n+1} + I_{n-1} &= \int_0^\beta u^{n-1}\left(u^2+1\right)\mathrm{d}x = 2\int_0^\beta u^{n-1}\frac{du}{dx}\,\mathrm{d}x = \frac{2}{n}\left[u^n\right]_0^\beta \\ &= \frac{2}{n}\left((\sec\beta + \tan\beta)^n - 1\right), \end{aligned}$$

since $u = 1$ at $x = 0$. Halving gives the required identity.

Ask yourself The identity involves I_{n+1} and I_{n-1}, but the inequality is about I_n. How can the middle term be squeezed in?

Compare u^2+1 with $2u$. We have $u^2+1-2u=(u-1)^2\geq 0$, and for $0<x\leq\beta$ we have $\sec x>1$ and $\tan x>0$, so $u>1$ and the inequality is strict there. Multiplying by $u^{n-1}>0$ gives $u^{n+1}+u^{n-1}>2u^n$ on $(0,\beta]$, and integrating (the integrands are continuous and the inequality is strict on an interval of positive length) gives

$$I_{n+1}+I_{n-1}>2I_n, \qquad \text{so} \qquad I_n<\tfrac{1}{2}\left(I_{n+1}+I_{n-1}\right)=\frac{1}{n}\left((\sec\beta+\tan\beta)^n-1\right).$$

(Equivalently: $u'-u=u(\sec x-1)>0$ for $x>0$, so $I_n=\int_0^\beta u^{n-1}u\,\mathrm{d}x<\int_0^\beta u^{n-1}u'\,\mathrm{d}x$. This view is the one that carries over to part (ii).)

Part (ii)

Write $c=\cos\beta$, so $0<c<1$, and let $v=c\sec x+\tan x$. Then

$$v'=\frac{\mathrm{d}v}{\mathrm{d}x}=c\sec x\tan x+\sec^2 x.$$

At the endpoints, $v(0)=c=\cos\beta$ and $v(\beta)=1+\tan\beta$, which are exactly the two numbers in the target. So the bound is $\frac{1}{n}[v^n]_0^\beta=\int_0^\beta v^{n-1}v'\,\mathrm{d}x$, and it is enough to show that $v<v'$ throughout $[0,\beta]$.

> **Ask yourself** In part (i), u^2+1 was exactly $2u'$. What is the corresponding relation for v?

Repeating the calculation of part (i),

$$v^2+1=c^2\sec^2 x+2c\sec x\tan x+\tan^2 x+1=(1+c^2)\sec^2 x+2c\sec x\tan x,$$

so

$$2v'-(v^2+1)=(1-c^2)\sec^2 x=\sin^2\beta\,\sec^2 x.$$

It is no longer an equality, but the leftover term is positive. Combining with $v^2+1-2v=(v-1)^2$,

$$2(v'-v)=(v-1)^2+\sin^2\beta\,\sec^2 x>0,$$

because $\sin\beta\neq 0$ for $0<\beta<\frac{\pi}{2}$. Hence $v<v'$ for every $x\in[0,\beta]$.

Finally $v>0$ on $[0,\beta]$ (as $c>0$, $\sec x>0$ and $\tan x\geq 0$), so $v^{n-1}>0$ and $v^n=v^{n-1}v<v^{n-1}v'$. Integrating over $[0,\beta]$, for $n\geq 1$,

$$J_n=\int_0^\beta v^n\,\mathrm{d}x<\int_0^\beta v^{n-1}\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x=\frac{1}{n}\Big[v^n\Big]_0^\beta=\frac{1}{n}\left((1+\tan\beta)^n-\cos^n\beta\right),$$

as required.

Checks

- With $n=1$ the identity in (i) says $\frac{1}{2}(I_2+\beta)=\sec\beta+\tan\beta-1$. For $\beta=1$ a numerical integration gives both sides as $2.40822\ldots$.

- For $\beta = 1$, $n = 3$: numerically $I_3 \approx 8.774$ against the bound 12.863, and $J_3 \approx 3.326$ against the bound 5.523.
- Putting $\cos\beta = 1$ formally in (ii) turns J_n into I_n and the leftover term $\sin^2\beta \sec^2 x$ disappears: part (ii) really is a perturbation of part (i).

Common slips

- Forgetting to check positivity: multiplying $v < v'$ by v^{n-1} needs $v^{n-1} > 0$, which is why $v > 0$ matters.
- Claiming only $\leq$: the strict inequality needs $u > 1$ (or $\sin\beta \neq 0$) on an interval, not just at a point.
- Evaluating $v(\beta)$ carelessly: $\sec\beta\cos\beta = 1$, which is where the $1 + \tan\beta$ comes from.

Beyond STEP

The argument "if $0 < f \leq g$ pointwise, with strict inequality somewhere, then $\int f < \int g$" is the monotonicity of the integral, which an analysis course proves carefully from the definition of the Riemann integral. Comparing an awkward integrand with an exact derivative is a standard way to bound integrals and sums, and it reappears in estimates such as the integral test. Abbott's *Understanding Analysis* covers integration and these comparison arguments rigorously.

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